

# COMPREHENSIVE\_COMPLETION\_PLAN

## HelixCode Comprehensive Completion Plan

### Executive Summary

This document provides a complete audit of unfinished work in the HelixCode project and a detailed phased implementation plan to achieve: - 100% test coverage with NO skipped tests - Complete documentation for all modules - Full user manuals and tutorials - Complete video course production - Updated website content

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## PART 1: UNFINISHED WORK AUDIT

### 1.1 Code Implementation Gaps

**Critical: Placeholder Implementations (15+ locations)**

| File                                     | Line | Issue                                       | Priority |
|------------------------------------------|------|---------------------------------------------|----------|
| internal/cognee/performance_optimizer.go | 893  | getCPUUsage() returns 0.0 placeholder       | HIGH     |
| internal/cognee/performance_optimizer.go | 898  | getGPUUsage() returns 0.0 placeholder       | HIGH     |
| internal/memory/cognee_integration.go    | 114  | Placeholder for Cognee API (StoreMemory)    | HIGH     |
| internal/memory/cognee_integration.go    | 133  | Placeholder for Cognee API (RetrieveMemory) | HIGH     |
| internal/memory/cognee_integration.go    | 156  | Placeholder for Cognee API (GetContext)     | HIGH     |
| internal/memory/cognee_integration.go    | 176  | Placeholder for Cognee API (GetSystemInfo)  | HIGH     |

| File                                   | Line       | Issue                                                       | Priority |
|----------------------------------------|------------|-------------------------------------------------------------|----------|
| internal/memory/cognee_integration.go  | 193        | Placeholder for Cognee API (GetOptimizationRecommendations) | HIGH     |
| internal/memory/cognee_integration.go  | 216        | Placeholder for optimization application                    | HIGH     |
| internal/memory/cognee_integration.go  | 229        | Placeholder for health check                                | HIGH     |
| internal/workflow/planmode/executor.go | 367        | executeFileOperation() placeholder                          | MEDIUM   |
| internal/workflow/planmode/executor.go | 388        | executeCodeGeneration() placeholder                         | MEDIUM   |
| internal/workflow/planmode/executor.go | 394        | executeCodeAnalysis() placeholder                           | MEDIUM   |
| internal/workflow/planmode/executor.go | 400        | executeValidation() placeholder                             | MEDIUM   |
| internal/workflow/planmode/executor.go | 406        | executeTesting() placeholder                                | MEDIUM   |
| internal/llm/usage_analytics.go        | 566-583    | Placeholder analytics data                                  | MEDIUM   |
| internal/config/config.go              | 1464, 1482 | Placeholder save implementations                            | LOW      |

## Mock Implementations Needing Real Integration

| Package                                   | Component         | Status          | Action Required              |
|-------------------------------------------|-------------------|-----------------|------------------------------|
| memory/providers/faiss_provider.go        | Vector search     | Mock            | Integrate real FAISS library |
| memory/providers/mem0_provider.go         | Memory operations | Stub            | Implement Mem0 API calls     |
| memory/providers/character_ai_provider.go | AI interface      | Mock            | Implement CharacterAI API    |
| memory/providers/weaviate_provider.go     | Backup/Restore    | Not implemented | Use Weaviate backup API      |
| memory/providers/zep_provider.go          | Embedding search  | Placeholder     | Implement real search        |

## TODO/FIXME Items in Code

| Location                              | Content                            | Category        |
|---------------------------------------|------------------------------------|-----------------|
| internal/workflow/<br>executor.go:633 | Go template<br>implementation      | Code Generation |
| internal/workflow/<br>executor.go:644 | Node.js template<br>implementation | Code Generation |
| internal/workflow/<br>executor.go:673 | Rust template<br>implementation    | Code Generation |
| internal/server/<br>handlers.go:286   | User authentication<br>placeholder | Security        |

## Missing API Endpoints (E2E Tests Expecting)

| Endpoint                       | Test File                                              | Status             |
|--------------------------------|--------------------------------------------------------|--------------------|
| /api/v1/llm/<br>providers      | tests/e2e/phase2/<br>real_server_test.go:213           | Not<br>implemented |
| /api/v1/<br>server/info        | tests/e2e/phase2/<br>real_server_test.go:281           | Not<br>implemented |
| /api/v1/<br>metrics            | tests/e2e/phase2/<br>real_server_test.go:298           | Not<br>implemented |
| /api/v1/<br>memory/<br>systems | tests/e2e/phase3/<br>production_validation_test.go:141 | Not<br>implemented |

## 1.2 Skipped Tests Analysis (MUST BE FIXED)

### Database-Related Skips (19 tests)

**Location:** internal/task/manager\_test.go

Tests skipped due to “Checkpoint tests require real database”: 1. TestCheckpointCreation 2. TestCheckpointRestore 3. TestCheckpointWithDependencies 4. TestCheckpointPersistence 5. TestCheckpointRecovery 6. TestTaskManagerWithDB 7. TestTaskPersistence 8. TestTaskRetrieval 9. TestTaskUpdate 10. TestTaskDeletion 11. TestTaskDependencies 12. TestTaskCheckpointing 13. TestBulkTaskOperations 14. TestConcurrentTaskAccess 15.

TestSplitTask (requires SplitStrategy) 16-19. Additional database integration tests

**Solution:** Provide PostgreSQL container in test infrastructure.

## **Browser Tests (12+ tests)**

**Location:** internal/tools/browser/browser\_test.go

Tests skipped due to “Chrome not installed”: - TestBrowserNavigation -  
TestBrowserScreenshot - TestBrowserClick - TestBrowserType -  
TestBrowserWait - TestBrowserExtract - TestBrowserPDF -  
TestBrowserCookies - TestBrowserLocalStorage -  
TestBrowserExecuteScript - TestBrowserNetwork - TestBrowserMultiTab

**Solution:** Provide Chrome/Chromium in test container.

## **SSH/Worker Tests (3+ tests)**

**Location:** internal/worker/distributed\_manager\_test.go,  
ssh\_security\_test.go

Tests skipped due to “Requires SSH setup” or “Requires root privileges”: -  
TestDistributedWorkerManager - TestSSHWorkerPool - TestSSHSecurity

**Solution:** Provide SSH server container with test keys.

## **Network Tests (3 tests)**

**Location:** internal/discovery/broadcast\_test.go

Tests skipped due to “UDP multicast unreliable in test environment”: -  
TestBroadcastDiscovery - TestMulticastListener -  
TestNetworkAnnouncement

**Solution:** Use Docker network with multicast support.

## **Voice/Audio Tests**

**Location:** internal/tools/voice/voice\_test.go

Tests skipped due to “no devices available”: - TestDeviceManager\_ListDevices  
- TestVoiceRecording

**Solution:** Use virtual audio device in container (PulseAudio null sink).

## LLM Provider Integration Tests (35+ tests)

**Location:** `internal/llm/integration_test.go`,  
`local_providers_integration_test.go`

Tests conditionally skipped when providers unavailable: - Ollama tests (5+) - Llama.cpp tests (3+) - vLLM tests (3+) - LocalAI tests (3+) - FastChat tests (3+) - TextGen tests (3+) - LM Studio tests (3+) - Jan AI tests (2+) - KoboldAI tests (2+) - GPT4All tests (2+) - TabbyAPI tests (2+) - MLX tests (2+) - MistralRS tests (2+)

**Solution:** Provide Ollama container with test model for core tests.

## Cloud Provider Tests (8+ tests)

Tests skipped due to missing API keys: - AWS Bedrock tests - OpenAI tests - Anthropic tests - Gemini tests - Azure OpenAI tests

**Solution:** Provide mock server container OR require API keys in CI secrets.

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# 1.3 Application Completion Status

## Terminal UI (`applications/terminal_ui/`)

- **Status:** 70% complete
- **Missing:**
  - Worker Management implementation
  - Project Management implementation
  - Session Management implementation
  - LLM Interaction implementation
  - Full Cognee integration
- **Tests:** Basic (134 lines) - needs expansion

## Desktop (`applications/desktop/`)

- **Status:** 80% complete
- **Missing:**
  - Projects tab (placeholder)
  - Sessions tab (placeholder)
  - LLM tab (placeholder)
  - Real worker data integration
- **Tests:** Theme-focused only (125 lines)

## **Aurora OS (applications/aurora\_os/)**

- **Status:** 75% complete
- **Missing:**
  - Projects/Sessions/LLM tabs
  - Real diagnostics implementation
  - Actual system metrics (currently simulated)
- **Tests:** Minimal (39 lines) - NEEDS SIGNIFICANT EXPANSION

## **Harmony OS (applications/harmony\_os/)**

- **Status:** 85% complete (BEST)
- **Missing:**
  - Worker registration
  - Task creation dialogs
  - Some Aurora features
- **Tests:** Most comprehensive (254 lines)

## **iOS (applications/ios/)**

- **Status:** 30% complete
- **Missing:**
  - Navigation controller
  - Task detail views
  - Project management
  - Worker management
  - Settings
  - Error handling
- **Tests:** NONE - NEEDS FULL TEST SUITE

## **Android (applications/android/)**

- **Status:** 20% complete
  - **Missing:**
    - Activity layouts
    - Navigation
    - All feature implementations
    - TaskAdapter implementation
  - **Tests:** NONE - NEEDS FULL TEST SUITE
-

## 1.4 Documentation Gaps

### Package Documentation: 100% COMPLETE

All 40 internal packages have README files.

### Minor Documentation Improvements Needed:

- `internal/focus/README.md` - Add more examples
  - `internal/logo/README.md` - Minimal, could expand
  - Cross-package references could be improved
- 

## 1.5 Video Course Status

### Course Platform: COMPLETE

- Player infrastructure: 100%
- Progress tracking: 100%
- Certificate system: 100%

### Video Content: NOT PRODUCED

- **Scripts Written:** 12+ Phase 3 scripts COMPLETE
  - **Actual Videos:** 0% - Using placeholder videos (BigBuckBunny, etc.)
  - **Videos Needed:** 28-32 videos across 4 courses
  - **Total Duration Needed:** ~10+ hours of content
- 

## 1.6 Website Status

### Main Website (`github_pages_website/`): 95% COMPLETE

- Version: v1.1.0 (November 2025)
- All sections implemented
- Mobile responsive
- Dark/Light theme

### Missing/Planned:

- Interactive demo playground
- Multi-language support (i18n)
- Real video content in courses section
- PDF certificate download (jsPDF needed)

## Secondary Website (website/): INCOMPLETE

- Only planning document exists
  - Hugo/Docusaurus not implemented
  - May be deprecated in favor of Github-Pages-Website
- 

# PART 2: TEST INFRASTRUCTURE FOR ZERO SKIPPED TESTS

## 2.1 Complete Docker Compose Test Environment

```
# docker-compose.full-test.yml  
# Complete test infrastructure ensuring NO tests are skipped
```

```
version: '3.8'
```

```
services:
```

```
  # =====
```

```
  # DATABASE SERVICES
```

```
  # =====
```

```
  postgres:
```

```
    image: postgres:16-alpine
```

```
    environment:
```

```
      POSTGRES_USER: helixcode
```

```
      POSTGRES_PASSWORD: ${HELIX_DATABASE_PASSWORD:?set  
        HELIX_DATABASE_PASSWORD in .env (run ./setup.sh, or  
        cp .env.example .env)}
```

```
      POSTGRES_DB: helixcode_test
```

```
    ports:
```

```
      - "5432:5432"
```

```
    healthcheck:
```

```
      test: ["CMD-SHELL", "pg_isready -U helixcode -d  
        helixcode_test"]
```

```
      interval: 5s
```

```
      timeout: 5s
```

```
      retries: 5
```

```
    volumes:
```

```
      - postgres_data:/var/lib/postgresql/data
```

```
    networks:
```

```
      - helixcode-test
```

```
  redis:
```

```
    image: redis:7-alpine
```

```
    ports:
```

```

    - "6379:6379"
healthcheck:
  test: ["CMD", "redis-cli", "ping"]
  interval: 5s
  timeout: 5s
  retries: 5
networks:
  - helixcode-test

# =====
# LLM PROVIDERS
# =====

ollama:
  image: ollama/ollama:latest
  ports:
    - "11434:11434"
  volumes:
    - ollama_data:/root/.ollama
  environment:
    - OLLAMA_HOST=0.0.0.0
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:11434/api/
      tags"]
    interval: 10s
    timeout: 10s
    retries: 10
  deploy:
    resources:
      reservations:
        devices:
          - driver: nvidia
            count: all
            capabilities: [gpu]
  networks:
    - helixcode-test

# Mock LLM server for cloud provider tests
mock-llm-server:
  build:
    context: ./tests/e2e/mocks
    dockerfile: Dockerfile.llm-mock
  ports:
    - "8090:8090"
  environment:
    - MOCK_OPENAI=true
    - MOCK_ANTHROPIC=true
    - MOCK_GEMINI=true
    - MOCK_AZURE=true

```

```

    - MOCK_BEDROCK=true
healthcheck:
  test: ["CMD", "curl", "-f", "http://localhost:8090/health"]
  interval: 5s
  timeout: 5s
  retries: 5
networks:
  - helixcode-test

# =====
# BROWSER TESTING
# =====

chrome:
  image: selenium/standalone-chrome:latest
  ports:
    - "4444:4444"
    - "7900:7900"
  shm_size: 2gb
  environment:
    - SE_NODE_MAX_SESSIONS=5
    - SE_NODE_SESSION_TIMEOUT=300
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:4444/status"]
    interval: 10s
    timeout: 10s
    retries: 5
  networks:
    - helixcode-test

# Chromedp-compatible Chrome
chromedp:
  image: chromedp/headless-shell:latest
  ports:
    - "9222:9222"
  networks:
    - helixcode-test

# =====
# SSH WORKER TESTING
# =====

ssh-server:
  build:
    context: ./tests/infrastructure
    dockerfile: Dockerfile.ssh-server
  ports:
    - "2222:22"
  environment:
    - SSH_USER=helixcode

```

```

    - SSH_PASSWORD=test_password
volumes:
    - ./tests/infrastructure/ssh_keys:/home/helixcode/.ssh:ro
healthcheck:
    test: ["CMD", "nc", "-z", "localhost", "22"]
    interval: 5s
    timeout: 5s
    retries: 5
networks:
    - helixcode-test

ssh-worker-1:
    build:
        context: ./tests/infrastructure
        dockerfile: Dockerfile.ssh-worker
    ports:
        - "2223:22"
    environment:
        - WORKER_ID=worker-1
        - SSH_USER=helixcode
    networks:
        - helixcode-test

ssh-worker-2:
    build:
        context: ./tests/infrastructure
        dockerfile: Dockerfile.ssh-worker
    ports:
        - "2224:22"
    environment:
        - WORKER_ID=worker-2
        - SSH_USER=helixcode
    networks:
        - helixcode-test

# =====
# AUDIO/VOICE TESTING
# =====
pulseaudio:
    image: thewierdnut/pulseaudio:latest
    environment:
        - PULSE_SERVER=unix:/run/pulse/native
    volumes:
        - pulse_socket:/run/pulse
    networks:
        - helixcode-test

# =====

```

```

# NETWORK TESTING (Multicast Support)
# =====
multicast-router:
  image: alpine:latest
  cap_add:
    - NET_ADMIN
  command: >
    sh -c "
      apk add --no-cache iproute2 &&
      sysctl -w net.ipv4.ip_forward=1 &&
      ip route add 224.0.0.0/4 dev eth0 &&
      tail -f /dev/null
    "
  networks:
    helixcode-test:
      ipv4_address: 172.28.0.254

# =====
# KNOWLEDGE GRAPH / MEMORY
# =====
cognee:
  image: topoteretes/cognee:latest
  ports:
    - "8000:8000"
  environment:
    - DATABASE_URL=postgresql://helixcode:${
      {HELIX_DATABASE_PASSWORD:?set HELIX_DATABASE_PASSWORD
      in .env (run ./setup.sh, or cp .env.example .env)}
      @postgres:5432/cognee
  depends_on:
    postgres:
      condition: service_healthy
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:8000/health"]
    interval: 10s
    timeout: 10s
    retries: 10
  networks:
    - helixcode-test

weaviate:
  image: semitechnologies/weaviate:1.24.1
  ports:
    - "8081:8080"
  environment:
    - QUERY_DEFAULTS_LIMIT=25
    - AUTHENTICATION_ANONYMOUS_ACCESS_ENABLED=true
    - PERSISTENCE_DATA_PATH=/var/lib/weaviate

```

```

    - DEFAULT_VECTORIZER_MODULE=none
volumes:
    - weaviate_data:/var/lib/weaviate
healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:8080/v1/.well-known/ready"]
    interval: 10s
    timeout: 10s
    retries: 5
networks:
    - helixcode-test

# =====
# HELIXCODE SERVER (Test Instance)
# =====
helixcode-server:
    build:
        context: .
        dockerfile: Dockerfile.test
    ports:
        - "8080:8080"
    environment:
        - HELIX_AUTH_JWT_SECRET=test-jwt-secret-for-testing-only
        - HELIX_DATABASE_HOST=postgres
        - HELIX_DATABASE_PORT=5432
        - HELIX_DATABASE_USER=helixcode
        - HELIX_DATABASE_PASSWORD=${HELIX_DATABASE_PASSWORD:?set HELIX_DATABASE_PASSWORD in .env (run ./setup.sh, or cp .env.example .env)}
        - HELIX_DATABASE_NAME=helixcode_test
        - HELIX_REDIS_HOST=redis
        - HELIX_REDIS_PORT=6379
        - OLLAMA_HOST=http://ollama:11434
        - CHROME_DEBUGGER_URL=ws://chromedp:9222
        - SSH_TEST_HOST=ssh-server
        - SSH_TEST_PORT=22
        - COGNEE_ENDPOINT=http://cognee:8000
        - WEAVIATE_ENDPOINT=http://weaviate:8080
        - MOCK_LLM_ENDPOINT=http://mock-llm-server:8090
    depends_on:
        postgres:
            condition: service_healthy
        redis:
            condition: service_healthy
        ollama:
            condition: service_healthy
    healthcheck:
        test: ["CMD", "curl", "-f", "http://localhost:8080/health"]

```

```

    interval: 10s
    timeout: 10s
    retries: 10
  networks:
    - helixcode-test

networks:
  helixcode-test:
    driver: bridge
    ipam:
      config:
        - subnet: 172.28.0.0/16

volumes:
  postgres_data:
  ollama_data:
  weaviate_data:
  pulse_socket:

```

## 2.2 Supporting Dockerfiles

### Dockerfile.ssh-server

```

# tests/infrastructure/Dockerfile.ssh-server
FROM alpine:3.19

RUN apk add --no-cache openssh-server openssh-client bash curl

RUN mkdir -p /var/run/sshd && \
  ssh-keygen -A && \
  adduser -D -s /bin/bash helixcode && \
  echo "helixcode:test_password" | chpasswd && \
  mkdir -p /home/helixcode/.ssh && \
  chown -R helixcode:helixcode /home/helixcode/.ssh

COPY ssh_keys/id_rsa.pub /home/helixcode/.ssh/authorized_keys
RUN chown helixcode:helixcode /home/helixcode/.ssh/
  authorized_keys && \
  chmod 600 /home/helixcode/.ssh/authorized_keys

RUN echo "PermitRootLogin no" >> /etc/ssh/sshd_config && \
  echo "PasswordAuthentication yes" >> /etc/ssh/sshd_config && \
  echo "PubkeyAuthentication yes" >> /etc/ssh/sshd_config

EXPOSE 22

CMD ["/usr/sbin/sshd", "-D", "-e"]

```

## Dockerfile.ssh-worker

```
# tests/infrastructure/Dockerfile.ssh-worker
FROM golang:1.24-alpine

RUN apk add --no-cache openssh-server bash curl git

RUN mkdir -p /var/run/sshd && \
    ssh-keygen -A && \
    adduser -D -s /bin/bash helixcode && \
    echo "helixcode:test_password" | chpasswd && \
    mkdir -p /home/helixcode/.ssh /home/helixcode/workspace

COPY ssh_keys/id_rsa.pub /home/helixcode/.ssh/authorized_keys
RUN chown -R helixcode:helixcode /home/helixcode

# Pre-install Go tools for worker
RUN go install golang.org/x/tools/gopls@latest && \
    go install github.com/golangci/golangci-lint/cmd/golangci-
        lint@latest

EXPOSE 22

CMD ["/usr/sbin/sshd", "-D", "-e"]
```

## Dockerfile.llm-mock

```
# tests/e2e/mocks/Dockerfile.llm-mock
FROM golang:1.24-alpine AS builder

WORKDIR /app
COPY . .
RUN go build -o mock-llm-server ./cmd/mock-server

FROM alpine:3.19
COPY --from=builder /app/mock-llm-server /usr/local/bin/
EXPOSE 8090
CMD ["mock-llm-server"]
```

## 2.3 Test Environment Setup Script

```
#!/bin/bash
# scripts/setup-full-test-env.sh

set -e

echo "=== HelixCode Full Test Environment Setup ==="
```

```

# 1. Generate SSH keys if not exists
if [ ! -f tests/infrastructure/ssh_keys/id_rsa ]; then
    echo "Generating SSH test keys..."
    mkdir -p tests/infrastructure/ssh_keys
    ssh-keygen -t rsa -b 4096 -f tests/infrastructure/ssh_keys/
        id_rsa -N "" -C "helixcode-test"
fi

# 2. Pull and start all containers
echo "Starting full test infrastructure..."
docker compose -f docker-compose.full-test.yml up -d

# 3. Wait for all services to be healthy
echo "Waiting for services to be healthy..."
services=("postgres" "redis" "ollama" "chrome" "ssh-server"
    "cognee" "weaviate" "mock-llm-server")
for service in "${services[@]}; do
    echo "Waiting for $service..."
    timeout 120 bash -c "until docker compose -f docker-
        compose.full-test.yml ps $service | grep -q healthy; do
        sleep 2; done"
done

# 4. Pull Ollama model for tests
echo "Pulling Ollama test model (llama2:7b)..."
docker compose
    -f docker-compose.full-test.yml exec ollama ollama pull
    llama2:7b

# 5. Initialize database schema
echo "Initializing database schema..."
docker compose -f docker-compose.full-test.yml exec helixcode-
    server /app/bin/helixcode migrate

# 6. Verify all services
echo "Verifying services..."
curl -sf http://localhost:5432 || echo "PostgreSQL: OK (port
    check via psql)"
curl -sf http://localhost:6379 || echo "Redis: OK (check via
    redis-cli)"
curl -sf http://localhost:11434/api/tags && echo "Ollama: OK"
curl -sf http://localhost:4444/status && echo "Selenium Chrome:
    OK"
curl -sf http://localhost:8080/health && echo "HelixCode Server:
    OK"
curl -sf http://localhost:8090/health && echo "Mock LLM Server:
    OK"

```

```
echo ""
echo "=== Full Test Environment Ready ==="
echo "Run tests with: make test-full"
```

## 2.4 Environment Variables for Full Testing

```
# .env.full-test
# Complete environment for zero-skip testing

# Database
HELIX_DATABASE_HOST=localhost
HELIX_DATABASE_PORT=5432
HELIX_DATABASE_USER=helixcode
HELIX_DATABASE_PASSWORD=CHANGE_ME_db_password
HELIX_DATABASE_NAME=helixcode_test
HELIX_DATABASE_ENABLED=true

# Redis
HELIX_REDIS_HOST=localhost
HELIX_REDIS_PORT=6379
HELIX_REDIS_ENABLED=true

# Authentication
HELIX_AUTH_JWT_SECRET=test-jwt-secret-for-testing-only-32chars

# LLM Providers
OLLAMA_HOST=http://localhost:11434
MOCK_LLM_ENDPOINT=http://localhost:8090
OPENAI_API_KEY=mock-openai-key
ANTHROPIC_API_KEY=mock-anthropic-key
GEMINI_API_KEY=mock-gemini-key

# Browser Testing
CHROME_DEBUGGER_URL=ws://localhost:9222
SELENIUM_URL=http://localhost:4444

# SSH Workers
SSH_TEST_HOST=localhost
SSH_TEST_PORT=2222
SSH_TEST_USER=helixcode
SSH_TEST_KEY_PATH=./tests/infrastructure/ssh_keys/id_rsa

# Memory Providers
COGNEE_ENDPOINT=http://localhost:8000
WEAVIATE_ENDPOINT=http://localhost:8081

# Test Configuration
SKIP_EXPENSIVE_TESTS=false
```

```
SKIP_HARDWARE_TESTS=false
TESTING_SHORT=false
CI_MODE=true
TEST_TIMEOUT=30m
PARALLEL_JOBS=4
```

---

# PART 3: PHASED IMPLEMENTATION PLAN

## Phase 1: Test Infrastructure (Week 1-2)

### 1.1 Container Infrastructure Setup

- ☐ Create `docker-compose.full-test.yml` with all services
- ☐ Create `Dockerfile.ssh-server` for SSH testing
- ☐ Create `Dockerfile.ssh-worker` for distributed worker tests
- ☐ Create `Dockerfile.llm-mock` for cloud provider mocking
- ☐ Create SSH key generation scripts
- ☐ Create environment setup scripts
- ☐ Verify all containers work together

### 1.2 Mock Server Implementation

- ☐ Implement OpenAI-compatible mock server
- ☐ Implement Anthropic mock endpoints
- ☐ Implement Gemini mock endpoints
- ☐ Implement Azure OpenAI mock endpoints
- ☐ Implement AWS Bedrock mock endpoints

Add configurable response delays and errors

### 1.3 Test Configuration Updates

☐

Update `config/test-config.yaml` for container endpoints

☐

Create `.env.full-test` with all required variables

☐

Update CI/CD pipeline to use full test infrastructure

☐

Add health check verification in test setup

---

## Phase 2: Fix All Skipped Tests (Week 2-4)

### 2.1 Database Tests

☐

Update `internal/task/manager_test.go` to use containerized PostgreSQL

☐

Remove all “requires real database” skip conditions

☐

Add database connection retry logic in tests

☐

Verify all 19 checkpoint tests pass

### 2.2 Browser Tests

☐

Update `internal/tools/browser/browser_test.go` to use Selenium/chromedp container

☐

Configure Chrome remote debugging URL from environment

☐

Remove “Chrome not installed” skip conditions

☐

Verify all 12+ browser tests pass

### 2.3 SSH/Worker Tests

☐

Update `internal/worker/distributed_manager_test.go` to use SSH container

- ☐ Update `internal/worker/ssh_security_test.go` to use SSH container
- ☐ Configure SSH connection from environment variables
- ☐ Remove “requires SSH setup” skip conditions
- ☐ Verify all worker tests pass

## 2.4 Network Tests

- ☐ Configure Docker network with multicast support
- ☐ Update `internal/discovery/broadcast_test.go` to use test network
- ☐ Remove “flaky network test” skip conditions
- ☐ Verify all discovery tests pass

## 2.5 Voice/Audio Tests

- ☐ Configure PulseAudio virtual device in container
- ☐ Update `internal/tools/voice/voice_test.go` to use virtual device
- ☐ Remove “no devices available” skip conditions
- ☐ Verify voice tests pass (or document hardware requirement)

## 2.6 LLM Provider Tests

- ☐ Configure Ollama container with test model
  - ☐ Update integration tests to use mock server for cloud providers
  - ☐ Remove all “provider not available” skip conditions
  - ☐ Verify all 35+ LLM tests pass
-

## Phase 3: Implement Missing Code (Week 4-8)

### 3.1 Cognition Integration (HIGH PRIORITY)

- ☐ Implement `getCPUUsage()` in `performance_optimizer.go`
- ☐ Implement `getGPUUsage()` in `performance_optimizer.go`
- ☐ Implement `StoreMemory()` API call in `cognition_integration.go`
- ☐ Implement `RetrieveMemory()` API call
- ☐ Implement `GetContext()` API call
- ☐ Implement `GetSystemInfo()` API call
- ☐ Implement `GetOptimizationRecommendations()` API call
- ☐ Implement health check with real Cognition
- ☐ Add comprehensive tests for all Cognition functions

### 3.2 Workflow Plan Mode (MEDIUM PRIORITY)

- ☐ Implement `executeFileOperation()` with real file operations
- ☐ Implement `executeCodeGeneration()` with LLM integration
- ☐ Implement `executeCodeAnalysis()` with code analysis tools
- ☐ Implement `executeValidation()` with validation checks
- ☐ Implement `executeTesting()` with test runner
- ☐ Add tests for all plan mode operations

### 3.3 Memory Providers

- ☐ Complete FAISS provider implementation (or document as optional)
- ☐

- ☐ Complete Mem0 provider implementation
- ☐ Complete CharacterAI provider implementation
- ☐ Implement Weaviate backup/restore using API
- ☐ Implement Zep embedding search
- ☐ Add tests for all memory providers

### 3.4 Usage Analytics

- ☐ Implement real provider usage tracking
- ☐ Implement success rate calculation
- ☐ Implement trend analysis
- ☐ Add tests for analytics

### 3.5 Missing API Endpoints

- ☐ Implement `/api/v1/llm/providers` endpoint
- ☐ Implement `/api/v1/server/info` endpoint
- ☐ Implement `/api/v1/metrics` endpoint
- ☐ Implement `/api/v1/memory/systems` endpoint
- ☐ Update E2E tests to use new endpoints

### 3.6 Code Generation Templates

- ☐ Implement Go template in workflow executor
- ☐ Implement Node.js template
- ☐ Implement Rust template
- ☐ Add tests for all templates

---

## Phase 4: Application Completion (Week 8-12)

### 4.1 Terminal UI

- ☐ Implement Worker Management view
- ☐ Implement Project Management view
- ☐ Implement Session Management view
- ☐ Implement LLM Interaction view
- ☐ Complete Cognee integration
- ☐ Add comprehensive tests (target: 500+ lines)

### 4.2 Desktop Application

- ☐ Implement Projects tab functionality
- ☐ Implement Sessions tab functionality
- ☐ Implement LLM tab functionality
- ☐ Connect to real worker data
- ☐ Add comprehensive tests

### 4.3 Aurora OS Application

- ☐ Implement Projects/Sessions/LLM tabs
- ☐ Implement real diagnostics
- ☐ Connect to real system metrics
- ☐ Expand test coverage (target: 200+ lines)

## 4.4 Harmony OS Application

- ☐ Implement worker registration
- ☐ Implement task creation dialogs
- ☐ Port relevant Aurora features
- ☐ Verify test coverage

## 4.5 iOS Application

- ☐ Implement navigation controller
- ☐ Implement task detail views
- ☐ Implement project management
- ☐ Implement worker management
- ☐ Implement settings
- ☐ Add proper error handling
- ☐ Create Swift test suite (XCTest)
- ☐ Target: 80%+ test coverage

## 4.6 Android Application

- ☐ Create activity layouts (XML)
- ☐ Implement navigation
- ☐ Implement all feature views
- ☐ Complete TaskAdapter implementation
- ☐ Add proper dependency injection
- ☐

- ☐ Create Kotlin test suite (JUnit/Espresso)
  - ☐ Target: 80%+ test coverage
- 

## Phase 5: Test Coverage Completion (Week 12-14)

### 5.1 Test Type Coverage

#### Unit Tests

- ☐ Ensure every package has `*_test.go` files
- ☐ Minimum 80% line coverage per package
- ☐ All public functions must have tests
- ☐ Mock all external dependencies

#### Integration Tests

- ☐ Test all database operations
- ☐ Test all Redis operations
- ☐ Test all API endpoints
- ☐ Test all LLM provider integrations
- ☐ Test all memory provider integrations

#### E2E Tests

- ☐ Complete challenge framework tests
- ☐ All 6 challenge definitions must pass
- ☐ Multi-interface tests (CLI, TUI, REST, WebSocket, Desktop, Mobile)
- ☐ Multi-provider tests

**Benchmark Tests**

- ☐ Add benchmarks for all critical paths
- ☐ Performance baselines established
- ☐ Regression detection configured

**Load Tests**

- ☐ Notification load tests
- ☐ API load tests
- ☐ Database load tests
- ☐ Worker pool load tests

**Security Tests**

- ☐ OWASP test coverage
- ☐ Authentication tests
- ☐ Authorization tests
- ☐ Input validation tests
- ☐ SQL injection prevention tests
- ☐ XSS prevention tests

**5.2 Coverage Targets**

| Package         | Current | Target |
|-----------------|---------|--------|
| internal/auth   | 85%     | 95%    |
| internal/llm    | 80%     | 95%    |
| internal/task   | 70%     | 95%    |
| internal/worker | 75%     | 95%    |

| Package           | Current | Target |
|-------------------|---------|--------|
| internal/workflow | 70%     | 95%    |
| internal/memory   | 65%     | 95%    |
| internal/cognee   | 60%     | 95%    |
| internal/tools    | 75%     | 95%    |
| internal/editor   | 90%     | 95%    |
| internal/server   | 80%     | 95%    |
| internal/database | 75%     | 95%    |
| applications/*    | 30%     | 80%    |

---

## Phase 6: Documentation Completion (Week 14-16)

### 6.1 Package Documentation

☐

Expand `internal/focus/README.md` with more examples

☐

Expand `internal/logo/README.md`

☐

Add cross-package references

☐

Document all sub-packages prominently

### 6.2 API Documentation

☐

Complete OpenAPI/Swagger specification

☐

Generate API reference from code

☐

Add request/response examples

☐

Document all error codes

### 6.3 Architecture Documentation

☐

Update architecture diagrams

☐

- ☐ Document data flow
- ☐ Document component interactions
- ☐ Document deployment architectures

## 6.4 Development Documentation

- ☐ Contributing guide updates
  - ☐ Code style guide
  - ☐ Testing guide
  - ☐ Debugging guide
- 

# Phase 7: User Manual Updates (Week 16-17)

## 7.1 Tutorial Updates

- ☐ Review and update Tutorial 1 (Web App)
- ☐ Review and update Tutorial 2 (Refactoring)
- ☐ Review and update Tutorial 3 (AI Providers)
- ☐ Review and update Tutorial 4 (Browser Automation)
- ☐ Review and update Tutorial 5 (Voice to Code)
- ☐ Review and update Tutorial 6 (Multi-File Edits)
- ☐ Review and update Tutorial 7 (Distributed Development)
- ☐ Review and update Tutorial 8 (Plan Mode)

## 7.2 New Tutorials

- ☐ Tutorial 9: Mobile Development (iOS/Android)
- ☐

- ☐ Tutorial 10: Custom Tool Development
- ☐ Tutorial 11: MCP Integration
- ☐ Tutorial 12: Enterprise Deployment

### 7.3 Reference Updates

- ☐ Update CLI reference
  - ☐ Update configuration reference
  - ☐ Update API reference
  - ☐ Update troubleshooting guide
- 

## Phase 8: Video Course Production (Week 17-24)

### 8.1 Pre-Production (Week 17-18)

- ☐ Set up recording environment (OBS/Camtasia)
- ☐ Configure microphone and audio
- ☐ Prepare demo projects
- ☐ Create slide templates
- ☐ Script review and finalization

### 8.2 Phase 3 Course Production (Week 18-20)

- ☐ Record Video 1: Phase 3 Overview (10 min)
- ☐ Record Video 2: Getting Started (10 min)
- ☐ Record Video 3: Session Fundamentals (10 min)
- ☐ Record Video 4: Advanced Sessions (10 min)
- ☐

- ☐ Record Video 5: Memory Basics (10 min)
- ☐ Record Video 6: Advanced Memory (10 min)
- ☐ Record Video 7: Persistence Basics (10 min)
- ☐ Record Video 8: Advanced Persistence (10 min)
- ☐ Record Video 9: Template Basics (10 min)
- ☐ Record Video 10: Advanced Templates (10 min)
- ☐ Record Video 11: Integration Patterns (10 min)
- ☐ Record Video 12: Advanced Workflows (10 min)

### **8.3 Introduction Course (Week 20-21)**

- ☐ Record Welcome to HelixCode (15 min)
- ☐ Record Installation and Setup (20 min)
- ☐ Record Your First Project (25 min)
- ☐ Record Understanding AI Providers (30 min)
- ☐ Record Distributed Computing (35 min)
- ☐ Record Automated Workflows (40 min)

### **8.4 Advanced Course (Week 21-22)**

- ☐ Record MCP Protocol (45 min)
- ☐ Record Mobile Clients (50 min)
- ☐ Record Custom Tools (55 min)
- ☐ Record Performance Optimization (45 min)

### **8.5 Production Deployment Course (Week 22-23)**

- ☐ Record Production Architecture (40 min)
- ☐ Record Docker and Kubernetes (50 min)
- ☐ Record Security Best Practices (45 min)
- ☐ Record Monitoring and Observability (30 min)

## 8.6 Post-Production (Week 23-24)

- ☐ Edit all videos
  - ☐ Add captions/subtitles
  - ☐ Create thumbnails
  - ☐ Upload to hosting
  - ☐ Replace placeholder URLs in course-data.js
  - ☐ Generate transcripts
  - ☐ Test playback on all platforms
- 

## Phase 9: Website Updates (Week 24-26)

### 9.1 Content Updates

- ☐ Update feature list with new capabilities
- ☐ Update provider list
- ☐ Update platform support matrix
- ☐ Update download links
- ☐ Update documentation links

## 9.2 Course Integration

☐

Replace placeholder video URLs

☐

Update course metadata

☐

Enable certificate downloads (implement jsPDF)

☐

Test all course functionality

## 9.3 New Features

☐

Add interactive demo playground

☐

Add live chat widget (optional)

☐

Implement multi-language support (i18n)

☐

Add PWA support

## 9.4 Quality Assurance

☐

Full link validation

☐

Cross-browser testing

☐

Mobile responsiveness testing

☐

Performance optimization

☐

SEO optimization

☐

Accessibility audit (WCAG 2.1 AA)

---

# PART 4: TEST TYPES AND FRAMEWORK

## 4.1 Supported Test Types (6 Types)

### Type 1: Unit Tests

- **Framework:** Go testing + testify
- **Location:** \*\_test.go files alongside source
- **Command:** go test -v ./...
- **Coverage:** go test -cover ./...

### Type 2: Integration Tests

- **Framework:** Go testing with build tags
- **Build Tag:** // +build integration
- **Location:** \*\_integration\_test.go files
- **Command:** go test -v -tags=integration ./...

### Type 3: E2E Tests

- **Framework:** Custom challenge framework
- **Location:** tests/e2e/
- **Components:**
  - Challenge definitions (tests/e2e/challenges/definitions/)
  - Core tests (tests/e2e/core/)
  - Mock services (tests/e2e/mocks/)
  - Orchestrator (tests/e2e/orchestrator/)
- **Command:** go run tests/e2e/challenges/cmd/runner/main.go

### Type 4: Benchmark Tests

- **Framework:** Go testing benchmarks
- **Location:** \*\_bench\_test.go or \*\_test.go with Benchmark\* functions
- **Command:** go test -bench=. -benchmem ./...
- **Count:** 129 benchmark functions

### Type 5: Load Tests

- **Framework:** Custom load testing
- **Location:** test/load/
- **Components:** Notification load tests, API load tests

## Type 6: Security Tests

- **Framework:** Custom OWASP-style tests
- **Location:** tests/security/
- **Coverage:** OWASP Top 10 vulnerabilities

## 4.2 Tests Bank Framework

### Structure

```
tests/
├── e2e/
│   ├── challenges/
│   │   ├── cmd/runner/main.go      # Test runner
│   │   ├── definitions/            # Challenge JSON files
│   │   ├── executor.go             # Challenge execution
│   │   ├── validator.go            # Code validation
│   │   ├── functional_validator.go # Runtime validation
│   │   ├── runtime_validator.go    # Execution validation
│   │   ├── usecase_validator.go    # Use case validation
│   │   ├── manager.go              # Challenge management
│   │   ├── types.go               # Type definitions
│   │   └── test-results/           # Generated projects
│   ├── core/                      # Core E2E tests
│   ├── mocks/                    # Mock services
│   └── orchestrator/              # Test orchestration
├── integration/                  # Integration tests
├── security/                     # Security tests
├── automation/                   # Automation tests
└── infrastructure/               # Test infrastructure
```

### Challenge Definitions

- notes-project.json
- url-shortener.json
- cli-task-manager.json
- ascii-art-generator.json
- json-validator-cli.json
- tic-tac-toe-tui.json

### Running Tests Bank

*# List all challenges*

```
go run tests/e2e/challenges/cmd/runner/main.go -list
```

*# Run single challenge*

```
go run tests/e2e/challenges/cmd/runner/main.go -challenge notes-project-001
```

```
# Run with specific interface
```

```
go run tests/e2e/challenges/cmd/runner/main.go -interfaces cli,tui
```

```
# Run with specific provider
```

```
go run tests/e2e/challenges/cmd/runner/main.go -provider ollama
```

```
# Run all challenges
```

```
go run tests/e2e/challenges/cmd/runner/main.go -all
```

---

## PART 5: MAKEFILE TARGETS

```
# Add to helix_code/Makefile
```

```
# =====
```

```
# FULL TEST INFRASTRUCTURE
```

```
# =====
```

```
.PHONY: test-infra-up test-infra-down test-infra-status
```

```
test-infra-up:
```

```
    @echo "Starting full test infrastructure..."
```

```
    docker compose -f docker-compose.full-test.yml up -d
```

```
    @echo "Waiting for services..."
```

```
    @sleep 30
```

```
    @echo "Pulling Ollama model..."
```

```
    docker compose -f docker-compose.full-test.yml exec ollama
```

```
        ollama pull llama2:7b || true
```

```
    @echo "Test infrastructure ready!"
```

```
test-infra-down:
```

```
    @echo "Stopping test infrastructure..."
```

```
    docker compose -f docker-compose.full-test.yml down -v
```

```
test-infra-status:
```

```
    docker compose -f docker-compose.full-test.yml ps
```

```
# =====
```

```
# COMPREHENSIVE TESTING
```

```
# =====
```

```
.PHONY: test-full test-unit test-integration test-e2e test-benchmark test-load test-security
```

```
test-full: test-infra-up
    @echo "Running ALL tests (no skips)..."
    set -a && source .env.full-test && set +a && \
    go test -v -count=1 -timeout=30m ./...
    go test -v -count=1 -tags=integration -timeout=30m ./...
    cd tests/e2e/challenges && go run cmd/runner/main.go -all
    @echo "All tests completed!"
```

```
test-unit:
    @echo "Running unit tests..."
    go test -v -count=1 -timeout=10m ./internal/...
```

```
test-integration: test-infra-up
    @echo "Running integration tests..."
    set -a && source .env.full-test && set +a && \
    go test -v -count=1 -tags=integration -timeout=20m ./...
```

```
test-e2e: test-infra-up
    @echo "Running E2E tests..."
    set -a && source .env.full-test && set +a && \
    cd tests/e2e/challenges && go run cmd/runner/main.go -all
```

```
test-benchmark:
    @echo "Running benchmark tests..."
    go test -bench=. -benchmem -run=^$ ./...
```

```
test-load: test-infra-up
    @echo "Running load tests..."
    go test -v -count=1 -timeout=30m ./test/load/...
```

```
test-security: test-infra-up
    @echo "Running security tests..."
    go test -v -count=1 -timeout=20m ./tests/security/...
```

```
# =====
# COVERAGE
# =====
```

```
.PHONY: coverage coverage-html coverage-report
```

```
coverage:
    @echo "Generating coverage report..."
    go test -coverprofile=coverage.out -covermode=atomic ./...
    go tool cover -func=coverage.out
```

```
coverage-html: coverage
    go tool cover -html=coverage.out -o coverage.html
    @echo "Coverage report: coverage.html"
```

### coverage-report:

```
@echo "Checking coverage thresholds..."
@go test -coverprofile=coverage.out ./... && \
COVERAGE=$(go tool cover -func=coverage.out | grep total |
    awk '{print $$3}' | sed 's/%//') && \
echo "Total coverage: $$COVERAGE%" && \
if [ $(echo "$$COVERAGE < 85" | bc -l) -eq 1 ]; then \
    echo "ERROR: Coverage below 85% threshold!"; \
    exit 1; \
fi
```

---

## PART 6: SUCCESS CRITERIA

### 6.1 Test Criteria

- ☐ ZERO skipped tests in any test run
- ☐ 95%+ code coverage for all internal packages
- ☐ 80%+ code coverage for applications
- ☐ All 6 test types passing
- ☐ All 6 challenge definitions passing
- ☐ No flaky tests

### 6.2 Code Criteria

- ☐ No placeholder implementations
- ☐ No TODO/FIXME items (except documented future enhancements)
- ☐ No mock implementations for production features
- ☐ All API endpoints implemented
- ☐ All applications feature-complete

### 6.3 Documentation Criteria

- ☐ README.md in every package
- ☐ Complete API documentation
- ☐ All tutorials verified and tested
- ☐ User manual covers all features

### 6.4 Video Course Criteria

- ☐ All 28-32 videos produced
- ☐ All placeholder URLs replaced
- ☐ Transcripts available
- ☐ Certificate generation working

### 6.5 Website Criteria

- ☐ All links valid
- ☐ Mobile responsive
- ☐ Courses section complete
- ☐ Documentation up-to-date
- ☐ WCAG 2.1 AA compliant

---

## APPENDIX A: ESTIMATED TIMELINE

| Phase                        | Duration | Start  | End    |
|------------------------------|----------|--------|--------|
| Phase 1: Test Infrastructure | 2 weeks  | Week 1 | Week 2 |

| Phase                           | Duration | Start   | End     |
|---------------------------------|----------|---------|---------|
| Phase 2: Fix Skipped Tests      | 2 weeks  | Week 2  | Week 4  |
| Phase 3: Implement Missing Code | 4 weeks  | Week 4  | Week 8  |
| Phase 4: Application Completion | 4 weeks  | Week 8  | Week 12 |
| Phase 5: Test Coverage          | 2 weeks  | Week 12 | Week 14 |
| Phase 6: Documentation          | 2 weeks  | Week 14 | Week 16 |
| Phase 7: User Manuals           | 1 week   | Week 16 | Week 17 |
| Phase 8: Video Production       | 7 weeks  | Week 17 | Week 24 |
| Phase 9: Website Updates        | 2 weeks  | Week 24 | Week 26 |

**Total Duration:** 26 weeks (6.5 months)

---

# APPENDIX B: RESOURCE REQUIREMENTS

## Hardware

- Development machine with 16GB+ RAM
- GPU for Ollama testing (recommended)
- Microphone for video recording
- Storage: 100GB+ for test artifacts

## Software

- Go 1.24+
- Docker & Docker Compose
- Chrome/Chromium
- OBS Studio or Camtasia (video recording)
- Audio editing software

## Services

- GitHub Actions (CI/CD)
  - Container registry
  - Video hosting platform
-

